

Jianfeng Lin

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EDUCATION

- **Georgia Institute of Technology (GT), Atlanta, GA** Aug. 2023 - Present
PhD in Quantitative Biosciences (QBioS)
Advisor: Daniel I. Goldman
- **Wuhan University (WHU), Wuhan, China** Sept. 2019 - June 2023
B.Eng in Mechanical Engineering
GPA: 3.64/4.0

EXPERIENCE

- **Complex Rheology and Biomechanics Lab at Georgia Tech** May 2024 - Present
Graduate Research Assistant
Advisor: Daniel I. Goldman
- **School of Physics, Georgia Tech** Aug. 2023 - May 2025
Graduate Teaching Assistant
PHYS 2211 - Intro Physics I: Fall 2023, Summer 2024, Spring 2025, Summer 2026
- **Dynamics Locomotion Group at Max Planck Institute for Intelligent Systems** July 2022 - Sept. 2023
Research Assistant
Remote
Advisor: Alexander Badri-Spröwitz
- **Exoskeleton Lab at Wuhan university** Aug. 2020 - June 2023
Undergraduate Research Assistant
Advisor: Zhao Guo

PATENTS AND PUBLICATIONS SELECTED PAPERS, S=IN SUBMISSION, J=JOURNAL, C=CONFERENCE, A=ABSTRACTS, P=PATENT

[†] Equally contributed

In Submission

- [S.3] Z. J. Xu, J. He, D. Aydan, M. Taylor, T. Wang, **J. Lin**, W. Dyer, D. I. Goldman. [A Robust Antenna Provides Tactile Feedback in a Multi-legged Robot](#).
- [S.2] B. Chong[†], T. Wang[†], K. Diaz, C. J. Pierce, E. Erickson, J. Whitman, Y. Deng, E. Flores, R. Fu, J. He, **J. Lin**, H. Lu, G. Sartoretti, H. Choset, D. I. Goldman. [The Omega Turn: A General Turning Template for Elongate Robots](#).
- [S.1] B. Chong, J. He, D. Irvine, T. Wang, E. Flores, D. Soto, **J. Lin**, Z. Xu, V. R. Nienhuser, G. Blekherman, D. I. Goldman. [Robust control for multi-legged elongate robots in noisy environments](#).

Journal Papers

- [J.3] **J. Lin**[†], T. Wang[†], B. Chong, M. Fernandez, Z. Xu, D. I. Goldman. [Optimal swimming with body compliance in an overdamped medium](#). *Physical Review Applied*, 2026, DOI: 10.1103/pbkh-py5h
- [J.2] J. He, B. Chong, **J. Lin**, Z. Xu, H. Bagheri, E. Flores, D. I. Goldman. [Probabilistic approach to feedback control enhances multi-legged locomotion on rugged landscapes](#). *IEEE Transactions on Robotics*, 2025, DOI: 110.1109/TRO.2025.3593133
- [J.1] **J. Lin**, J. Ke, R. Xiao, X. Jiang, M. Li, X. Xiao, Z. Guo. [Bioinspired bidirectional stiffening soft actuators enable versatile and robust grasping](#). *Soft Robotics*, 2024, DOI: 10.1089/soro.2022.0212

Conference Papers

- [C.3] M. Iaschi, B. Chong, T. Wang, **J. Lin**, J. He, D. Soto, Z. Xu, D. I. Goldman. [Addition of a peristaltic wave improves multi-legged locomotion performance on complex terrains](#). *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, DOI:10.1109/ICRA55743.2025.11127656
- [C.2] **J. Lin**, Z. Guo, A. Badri-Spröwitz. [Bird-inspired tendon coupling improves paddling efficiency by shortening phase transition times](#). *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, DOI: 10.1109/ICRA55743.2025.11128691
- [C.1] **J. Lin**, R. Xiao, Z. Guo. [Design and Stiffness Analysis of a Bio-Inspired Soft Actuator with Bi-Direction Tunable Stiffness Property](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, DOI: 10.1109/IROS55552.2023.10341730

Selected Conference Abstracts/Posters

- [A.1] J. Lin, Z. Guo, A. Badri-Spröwitz. Bird-inspired tendon coupling improves paddling efficiency by shortening phase transition times. *ICRA 2025 Workshop "Towards Agility and Robustness: Mechanical Intelligence in Robotics"*, 2025. **Best Design Award**

Patent

- [P.1] Z. Guo, J. Lin, R. Xiao. A rigid-soft coupling actuator with multi-directional variable stiffness properties and a manufacturing method. *China Patent*, CN114952937A, 2022

HONORS AND AWARDS

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- | | |
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| • Georgia Tech School of Physics Travel Grant (\$700) | Feb. 2026 |
| • Georgia Tech College of Sciences Graduate Career Connect Travel Grant (\$500) | Dec. 2025 |
| • QBioS Travel Award, covering the expense to SICB 2026 (\$1000) | Aug. 2025 |
| • Best Design Award in the ICRA 2025 Workshop, Mechanical Intelligence in Robotics (\$250) | May 2025 |
| • The Second Prize in the 10th National College Mechanical Innovation Competition of China | Aug. 2022 |
| • The Second Prize of Fujian Region of the 35th Chinese Physics Olympiad | Oct. 2018 |

ACADEMIC SERVICES

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| • Organizer of Robophysics Mini Boot Camp (NSF PoLS workshop) | 2024, 2025 |
| • Reviewer: Scientific Reports, IEEE RA-L | 2025 - Present |

PROFESSIONAL MEMBERSHIPS

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| • IEEE, Student Member | 2022 - Present |
| • Society of Comparative and Integrative Biology, Member | 2025 - Present |
| • American Physical Society, Member | 2025 - Present |

Last updated: May, 2026